Ioannis Mitliagkas

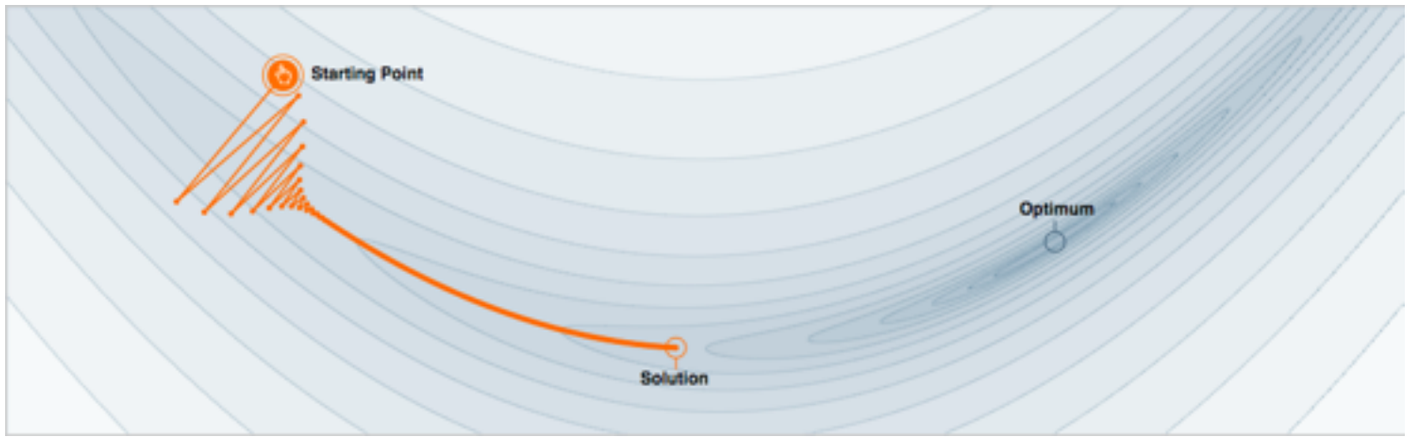
AN INTERESTING PROPERTY OF POLYAK'S MOMENTUM



GRADIENT DESCENT

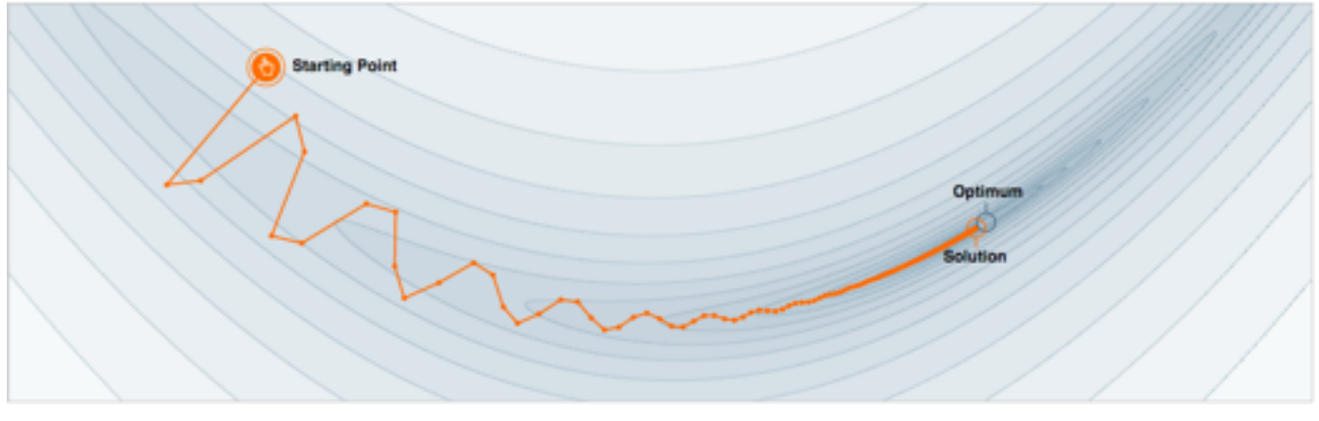
$$x_{t+1} = x_t - \alpha \nabla f(x_t)$$

Without momentum



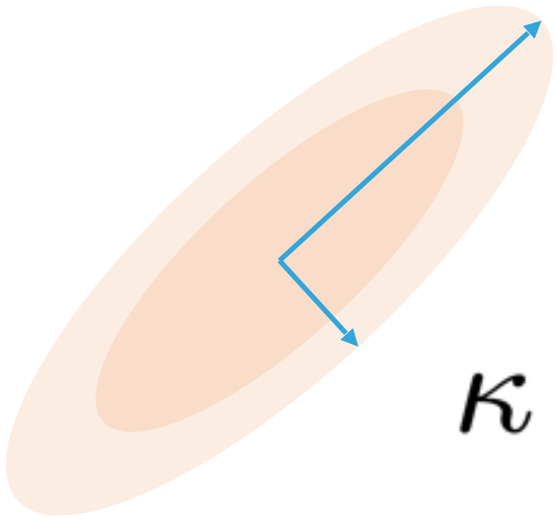
[Distill blog]

With momentum [Polyak, 1964]



CONDITION NUMBER

Dynamic range of curvatures, κ



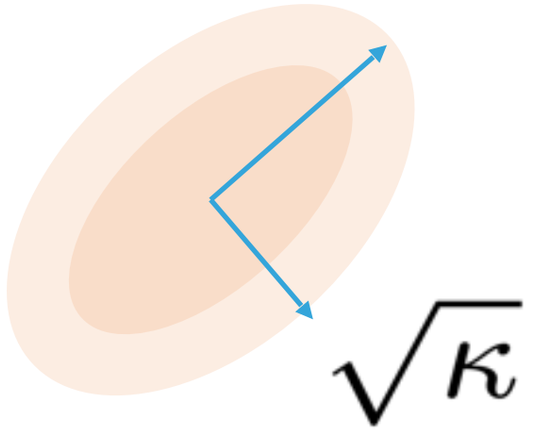
GRADIENT DESCENT ON STRONGLY CONVEX

Convergence rate $O(\frac{\kappa-1}{\kappa+1})$

GRADIENT DESCENT WITH MOMENTUM

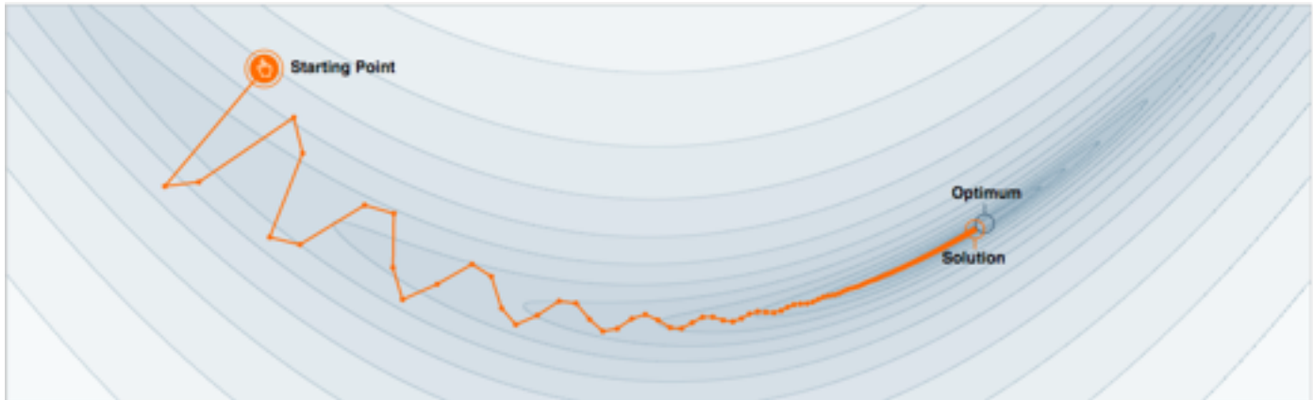
Dependence on κ changes

$$O(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1})^*$$



EFFECTIVELY IMPROVES THE CONDITION NUMBER

MOMENTUM ALGORITHM



[Distill blog]

Elegant, and very successful optimization method

Along with adaptive methods,
the **workhorse of modern machine learning**

BACKGROUND

GRADIENT DESCENT

$$x_{t+1} = x_t - \alpha \nabla f(x_t)$$

SCALAR QUADRATIC

Curvature

Minimize $$f(x) = \frac{h}{2}x^2$$

$$\begin{aligned} x_{t+1} &= x_t - \alpha \nabla f(x_t) \\ &= x_t - \alpha h x_t \\ &= (1 - \alpha h) x_t \end{aligned}$$ **LINEAR SYSTEM**

$$x_{t+1} = (1 - \alpha h)^t x_1$$

Rate of convergence

$$\rho = |1 - \alpha h|$$

RELAXATION PROPERTY

CONVERGENCE RATE

Rate $$\rho = |1 - \alpha h|$$

Convergence $$\|x_t - x^*\| \leq C \cdot \rho^t$$

(there exists C)

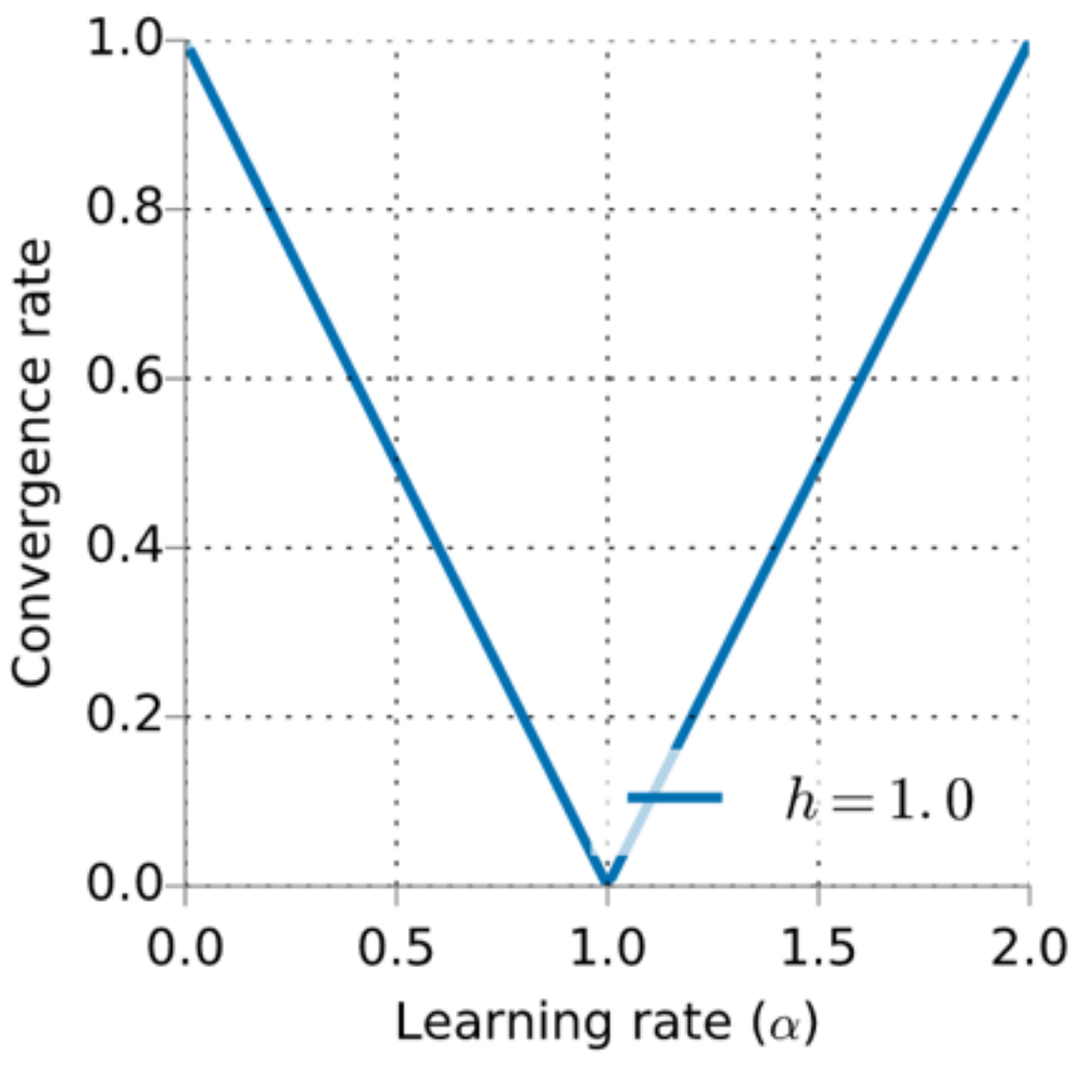
WE WANT THE RATE TO BE A SMALL NUMBER

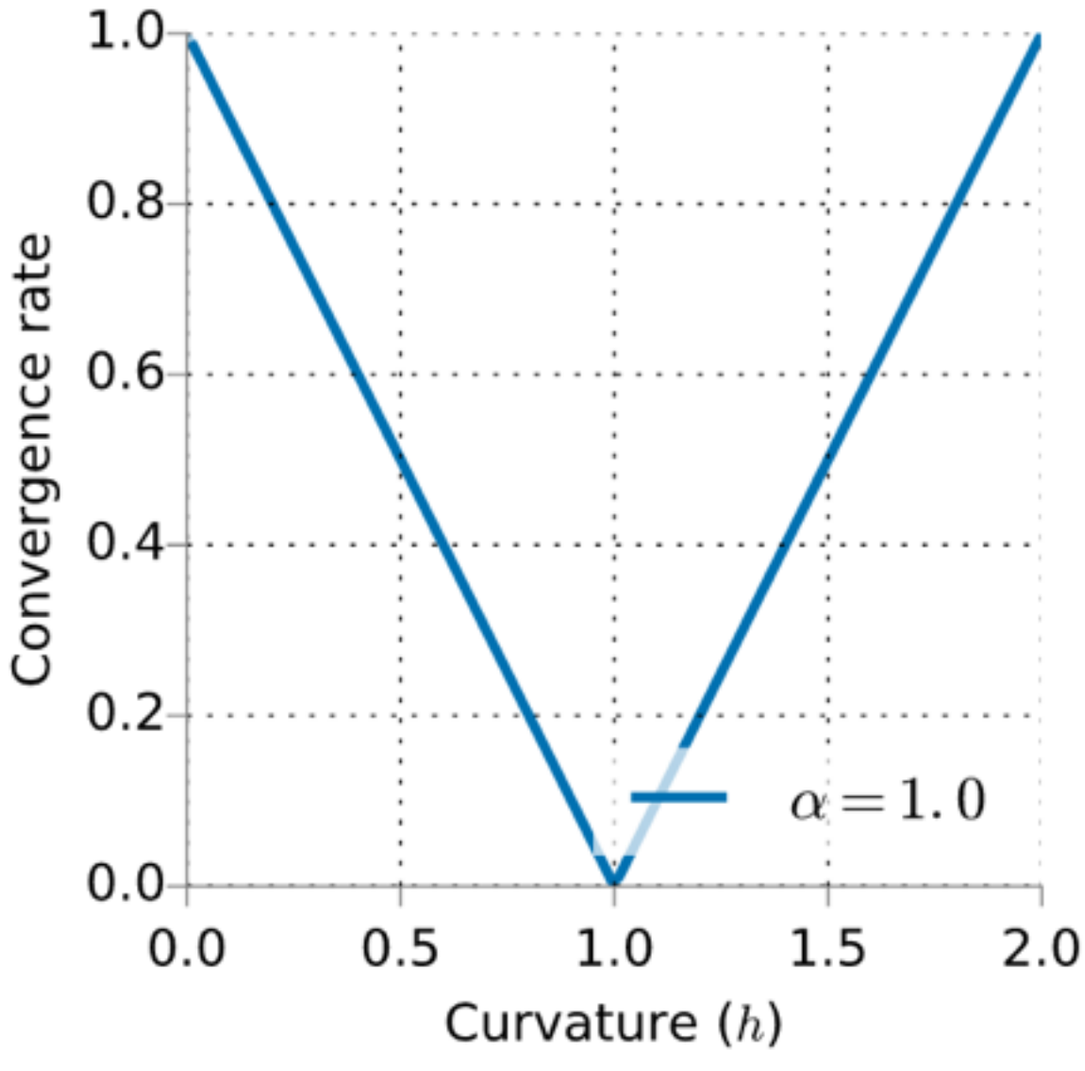
SCALAR QUADRATIC

$$f(x) = \frac{h}{2}x^2$$

Rate of convergence

$$\rho = |1 - \alpha h|$$





MULTIVARIATE QUADRATIC

WLOG, Hessian is diagonal
(why? separability along eigenbasis of H)

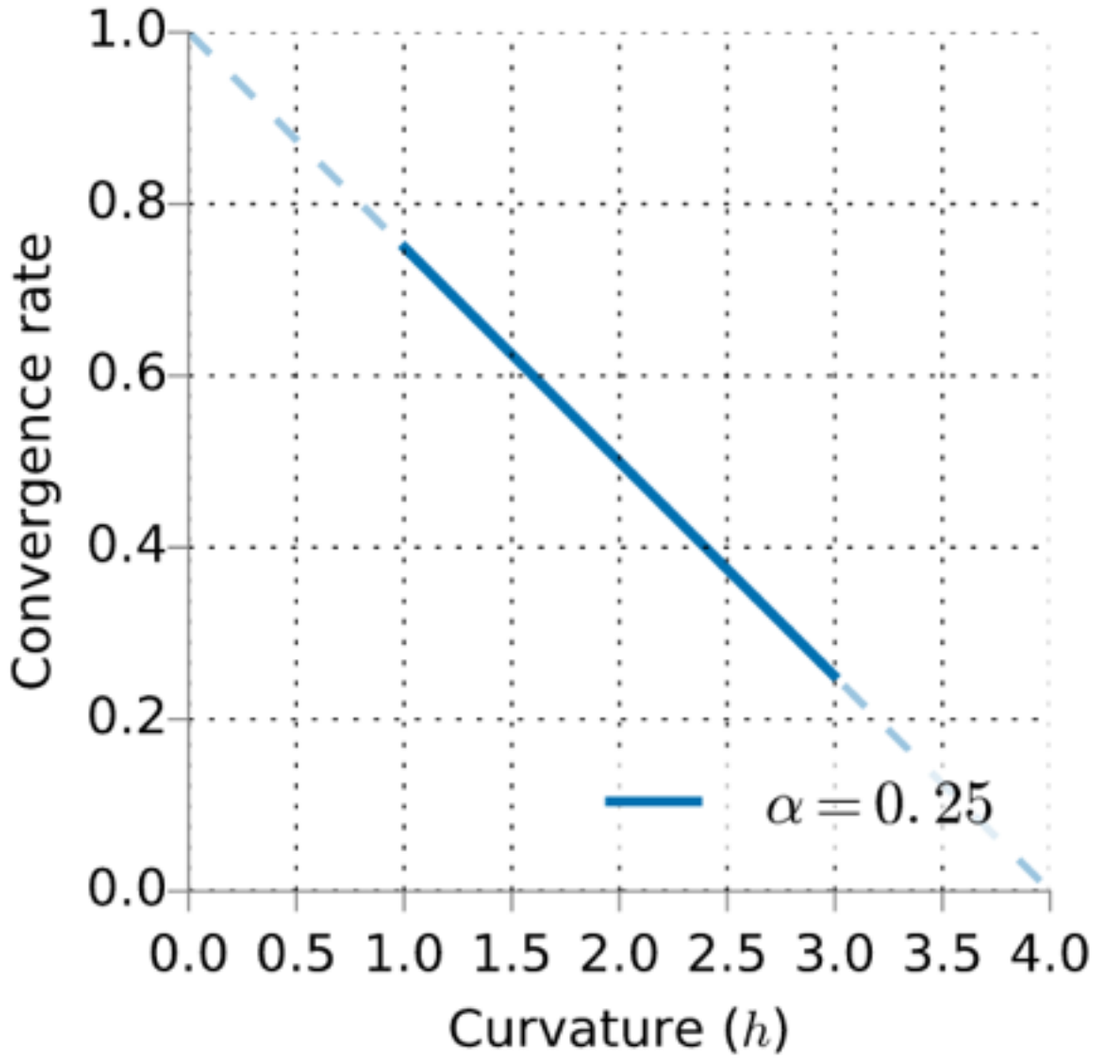
$$f(x) = \frac{1}{2} x^\top H x \qquad H = \begin{bmatrix} h_1 & 0 & 0 \\ 0 & h_2 & 0 \\ 0 & 0 & h_3 \end{bmatrix}$$

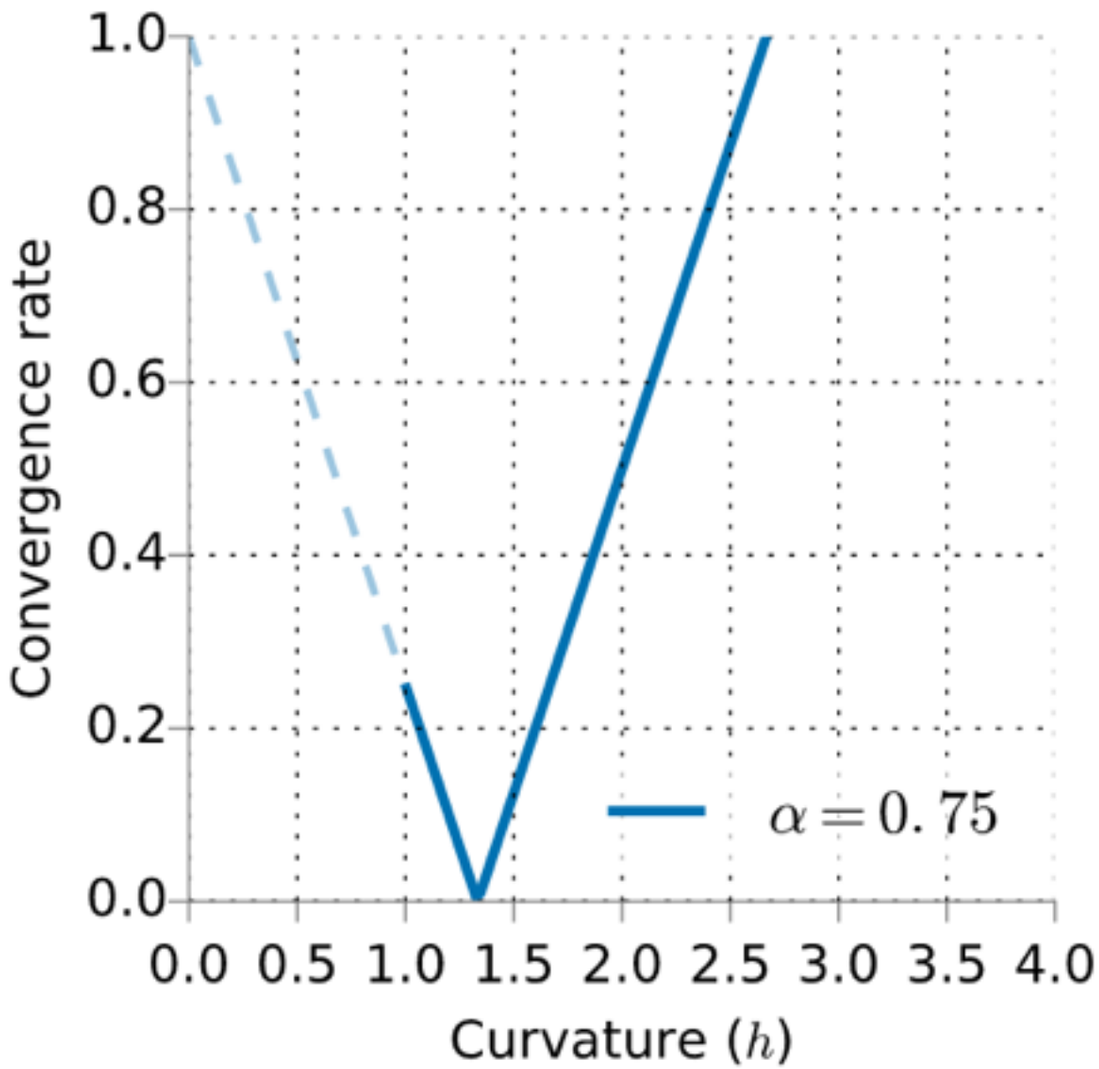
Analysis decomposes into scalar analyses of eigendirections

$$\begin{aligned} x_{t+1}(i) &= x_t(i) - \alpha h_i x_t(i) \\ &= (1 - \alpha h_i) x_t(i) \\ &= (1 - \alpha h_i)^t x_1(i) \end{aligned}$$

SLOWEST DIRECTION DOMINATES

$$H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

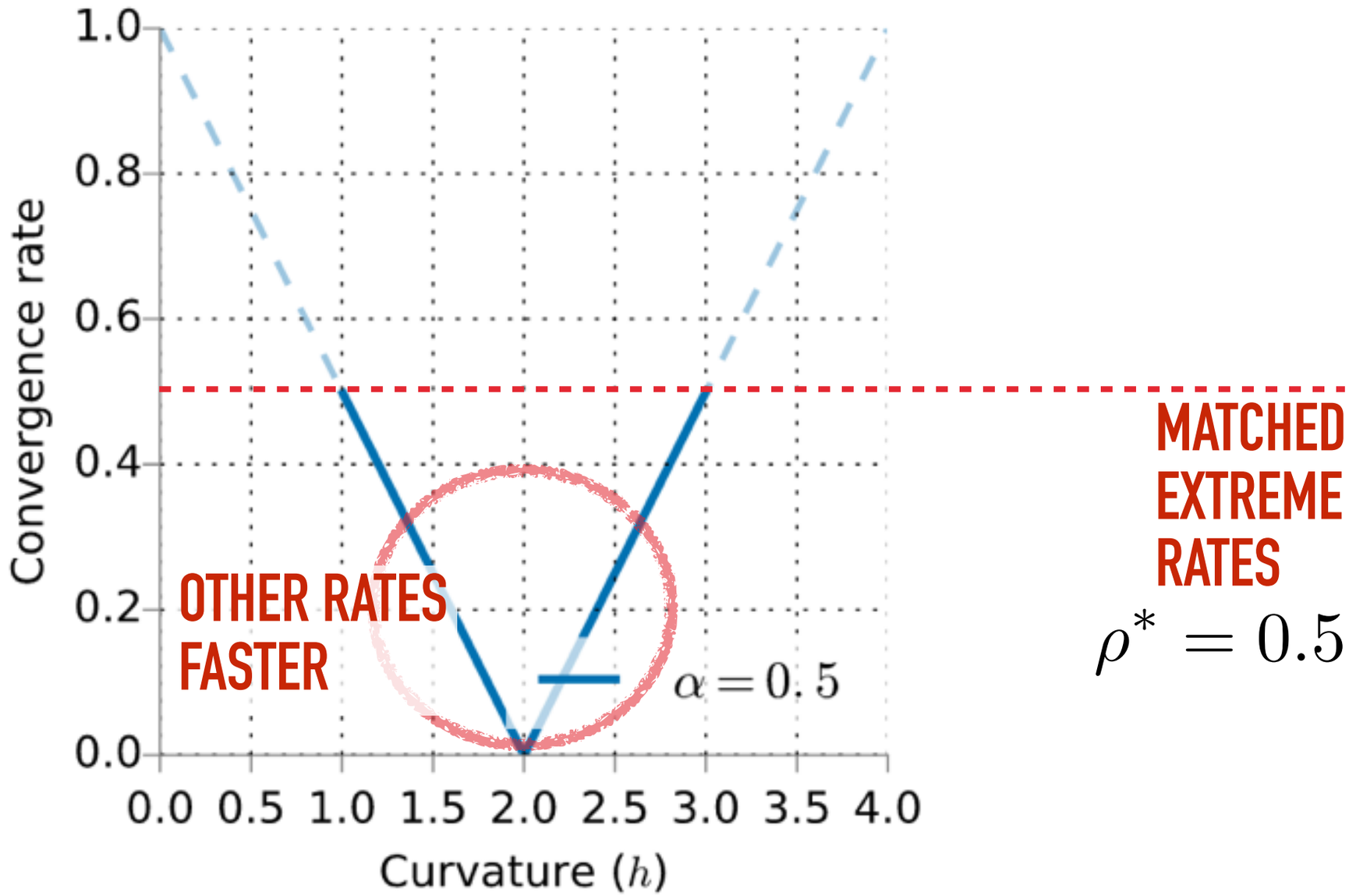




OPTIMAL LEARNING RATE

$$\alpha = 2/(h_{min} + h_{max}) = 0.5$$

$$H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$



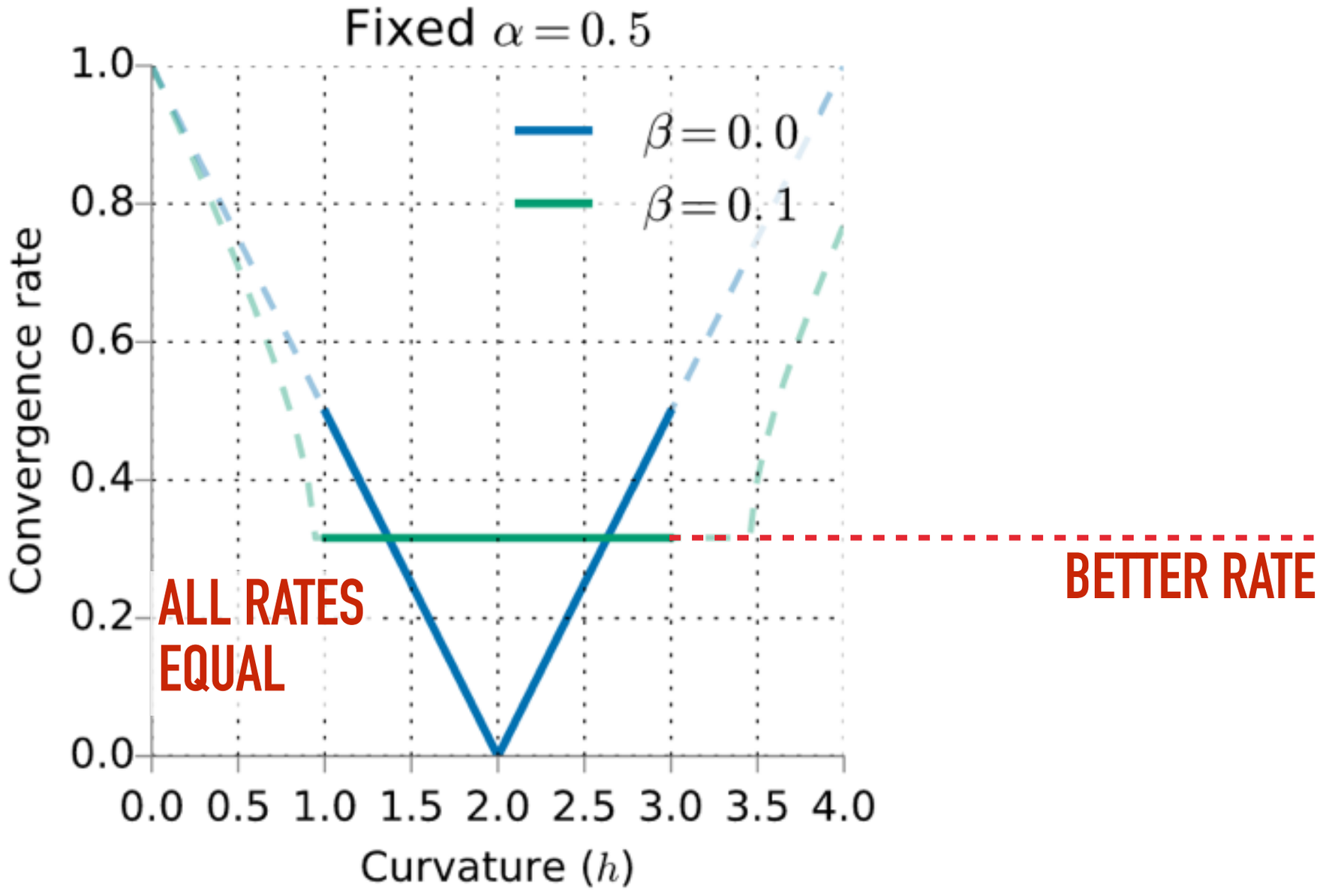
Polyak's momentum

(heavy ball method)

$$x_{t+1} = x_t - \alpha \nabla f(x_t) + \beta(x_t - x_{t-1})$$

PREVIEW

$$H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$



WHAT'S HAPPENING?

POLYAK'S MOMENTUM

Minimize $f(x) = \frac{h}{2}x^2$ (Curvature)

$$\begin{aligned} x_{t+1} &= x_t - \alpha \nabla f(x_t) + \beta(x_t - x_{t-1}) \\ &= x_t - \alpha h x_t + \beta(x_t - x_{t-1}) \\ &= (1 + \beta - \alpha h)x_t - \beta x_{t-1} \end{aligned}$$

CAN WE WRITE AS A LINEAR SYSTEM?

STATE SPACE AUGMENTATION

$$f(x) = \frac{h}{2}x^2$$

$$x_{t+1} = (1 + \beta - \alpha h)x_t - \beta x_{t-1}$$

$$\begin{bmatrix} x_{t+1} \\ x_t \end{bmatrix} = \begin{bmatrix} 1 - \alpha h + \beta & -\beta \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_t \\ x_{t-1} \end{bmatrix}$$

LINEAR OPERATOR, A

$$\begin{bmatrix} x_{t+1} \\ x_t \end{bmatrix} = A^t \begin{bmatrix} x_1 \\ x_0 \end{bmatrix}$$

CONVERGENCE RATE?

CONVERGENCE RATE

$$\begin{bmatrix} x_{t+1} \\ x_t \end{bmatrix} = A^t \begin{bmatrix} x_1 \\ x_0 \end{bmatrix}$$

RELAXATION?

Asymptotically, spectral radius of A gives rate

$$\left\| \begin{bmatrix} x_{t+1} \\ x_t \end{bmatrix} \right\| = O(\rho(A)^t)$$

MORE ATTENTION NEEDED FOR FINITE STEPS

$$\rho(A) = \max\{|\lambda_1(A)|, |\lambda_2(A)|\}$$

SPECTRUM OF MOMENTUM OPERATOR

$$A = \begin{bmatrix} 1 - \alpha h + \beta & -\beta \\ 1 & 0 \end{bmatrix}$$

First sign of 'robustness': $\lambda_1 \lambda_2 = \det(A) = \beta$

Two regions, depending on discriminant

$$\Delta = tr(A)^2 - 4det(A)$$

$\Delta \geq 0$

Two real eigenvalues

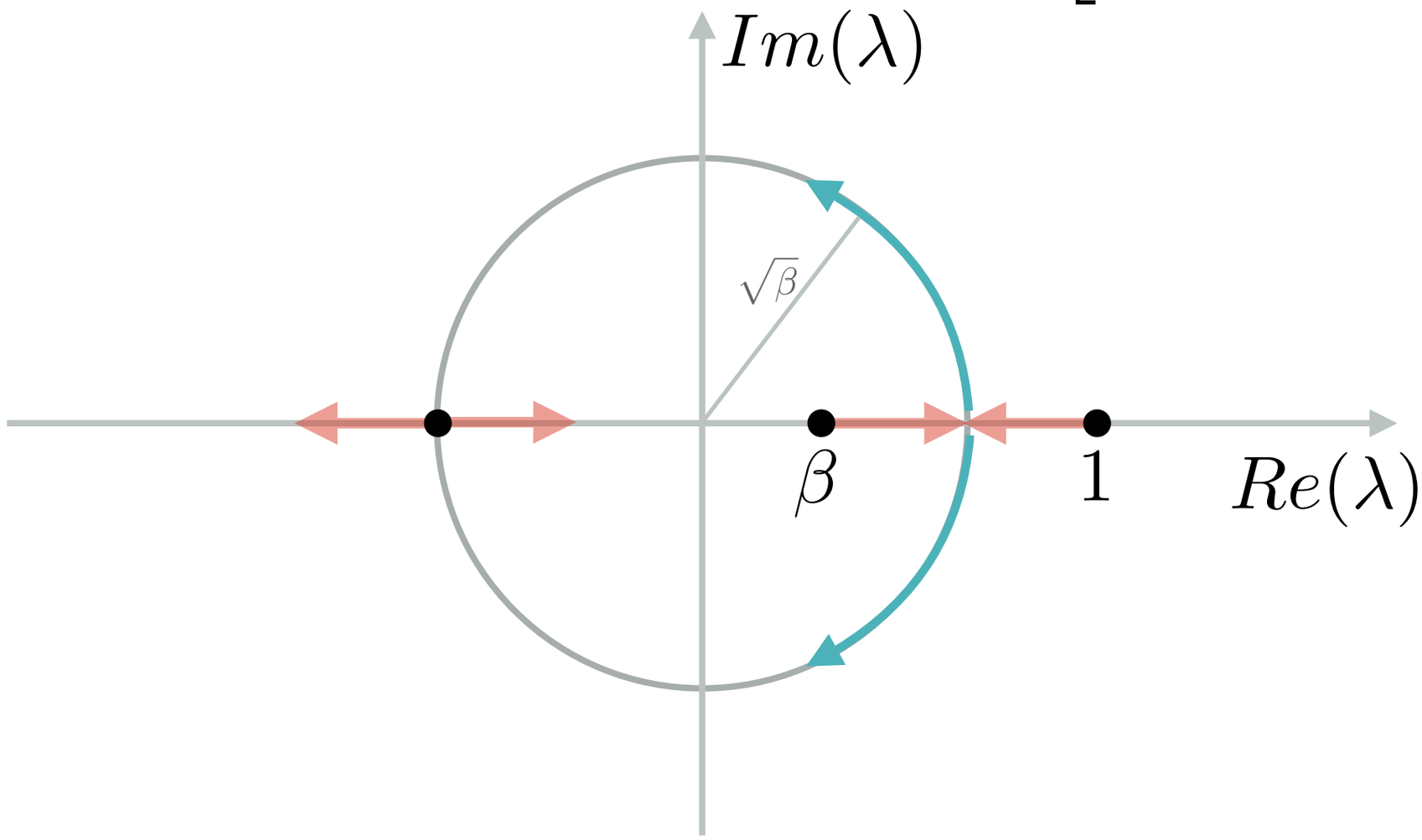
$\Delta < 0$

Two complex conjugate eigenvalues

$$\frac{(1 - \sqrt{\beta})^2}{h} \leq \alpha \leq \frac{(1 + \sqrt{\beta})^2}{h}$$

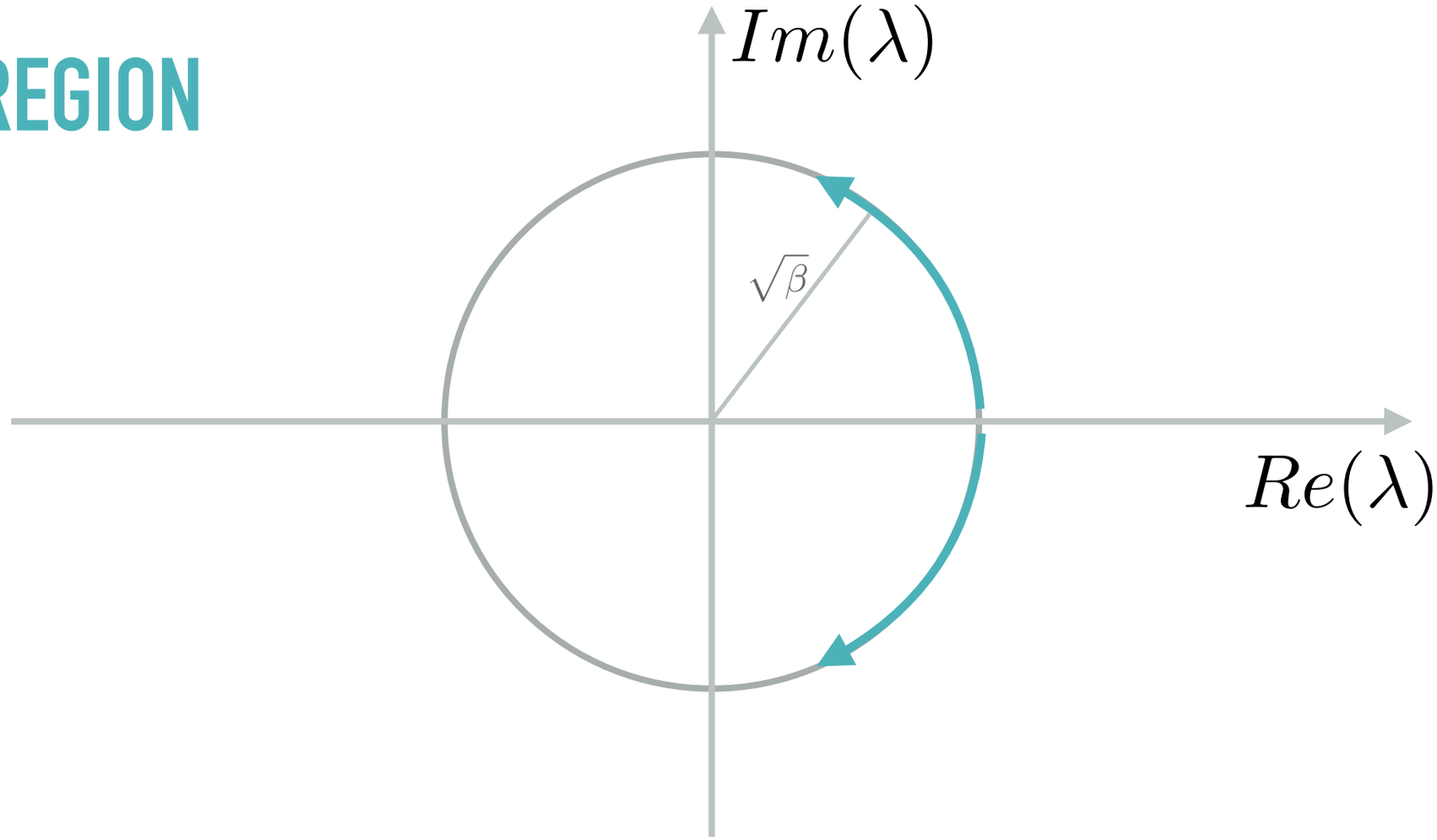
SPECTRUM OF MOMENTUM OPERATOR

$$A = \begin{bmatrix} 1 - \alpha h + \beta & -\beta \\ 1 & 0 \end{bmatrix}$$





ROBUST REGION





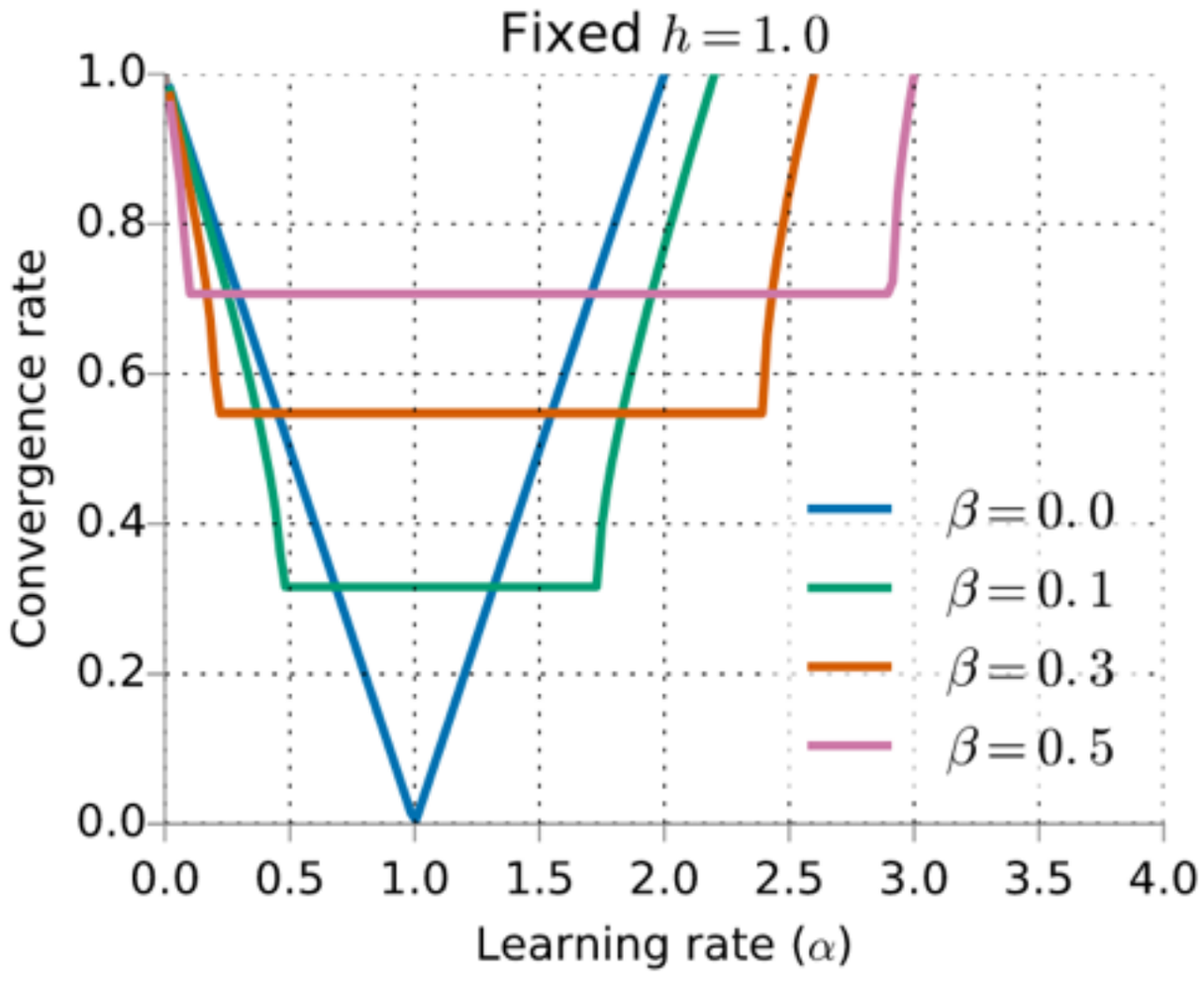
$$\frac{(1-\sqrt{\beta})^2}{h} \leq \alpha \leq \frac{(1+\sqrt{\beta})^2}{h}$$

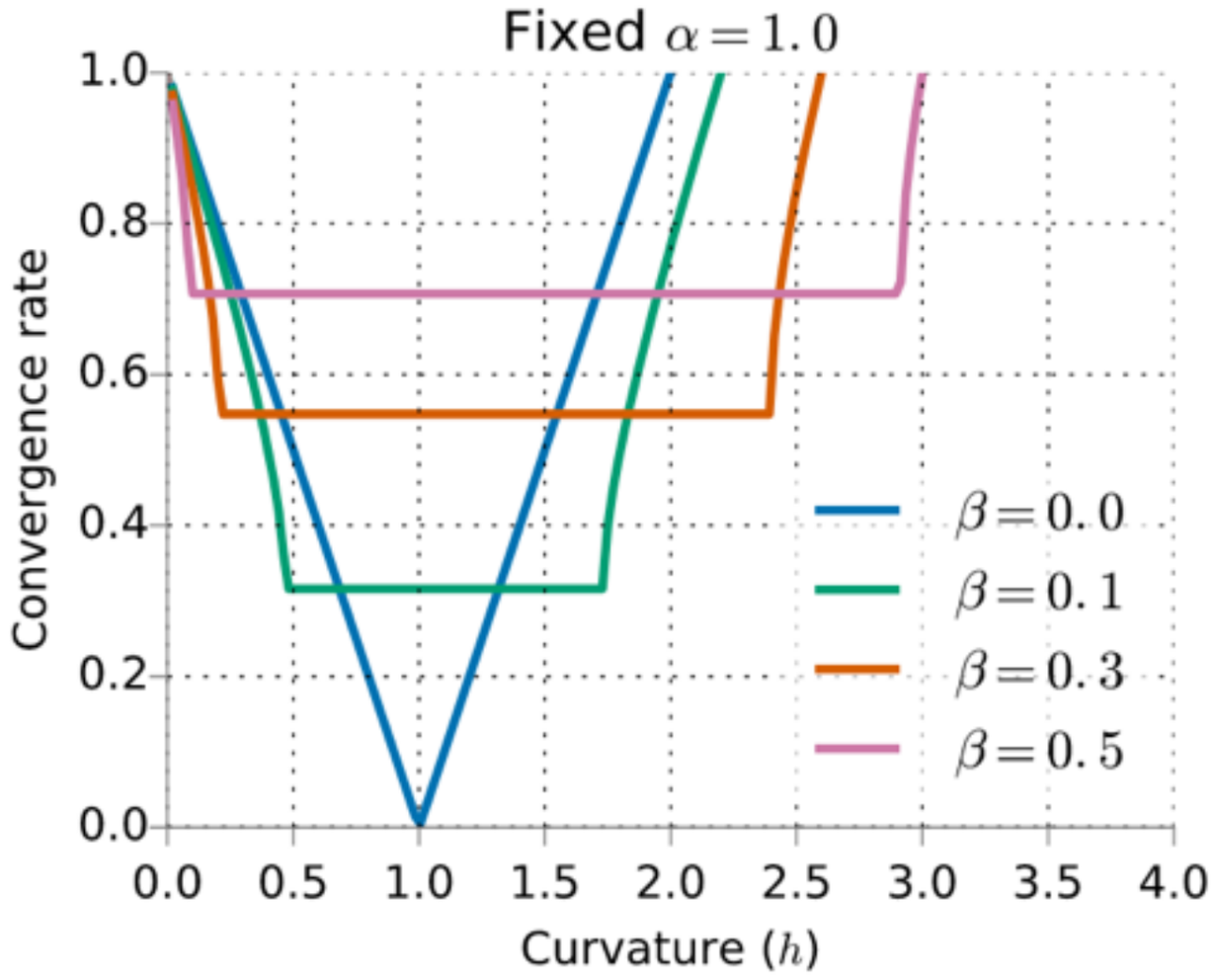
$$\rho(A) = \sqrt{\beta}$$

MOMENTUM, β, CONTROLS WIDTH OF ROBUST REGION

ROBUST REGION

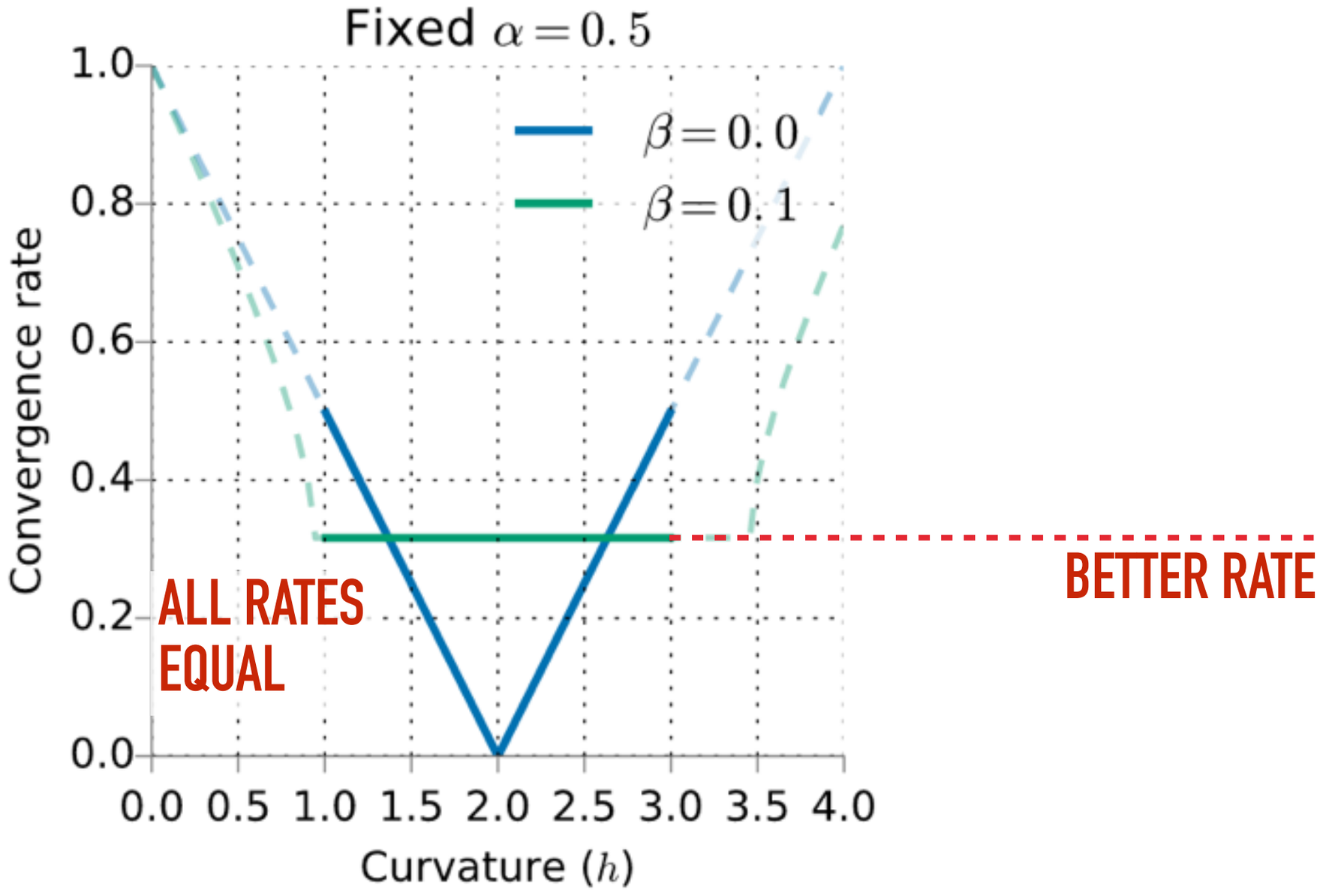
$$\frac{(1-\sqrt{\beta})^2}{h} \leq \alpha \leq \frac{(1+\sqrt{\beta})^2}{h} \qquad \rho(A) = \sqrt{\beta}$$



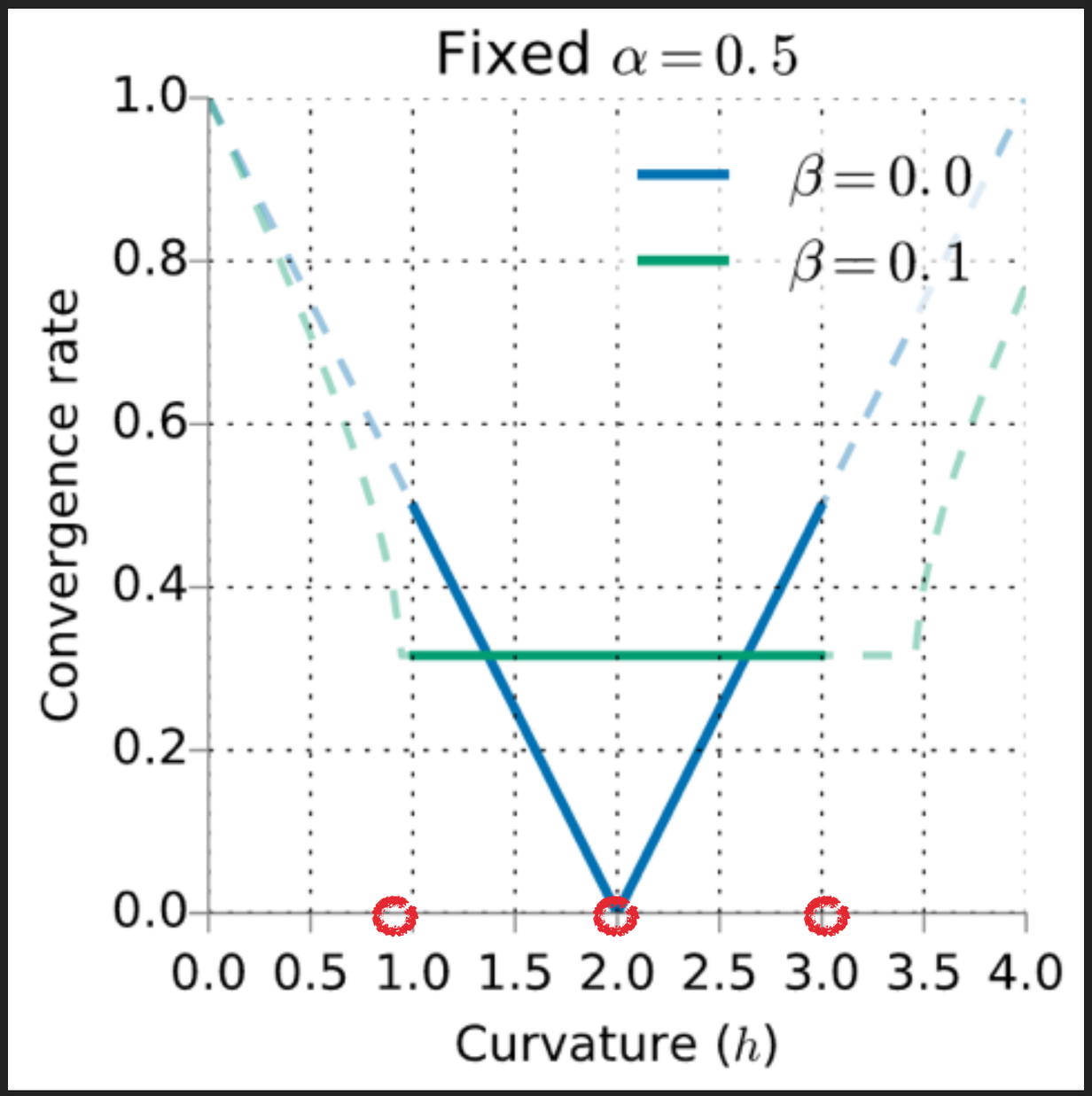


BACK TO EXAMPLE

$$H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

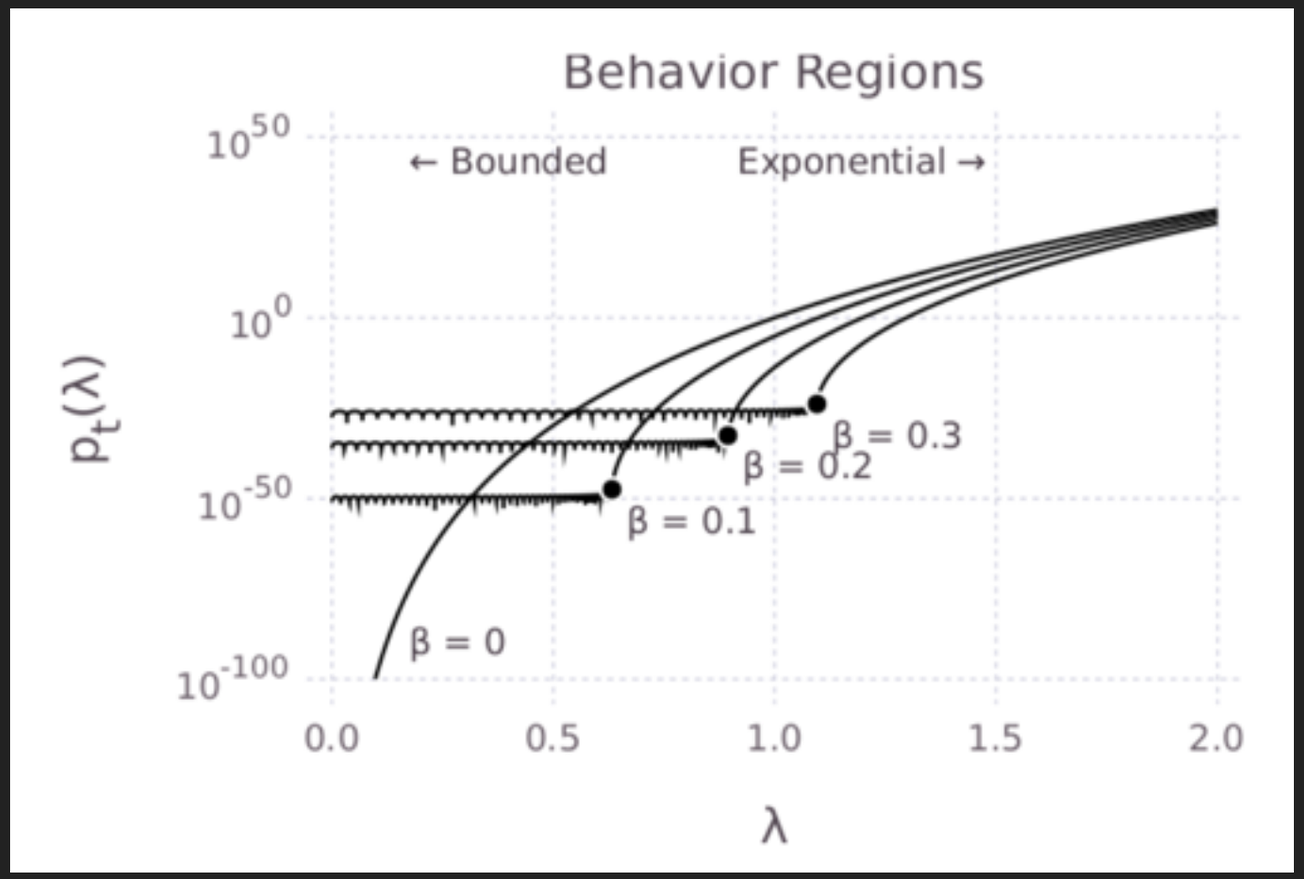


THEME: TUNE LINEAR OPERATOR TO MANIPULATE EIGENVALUES

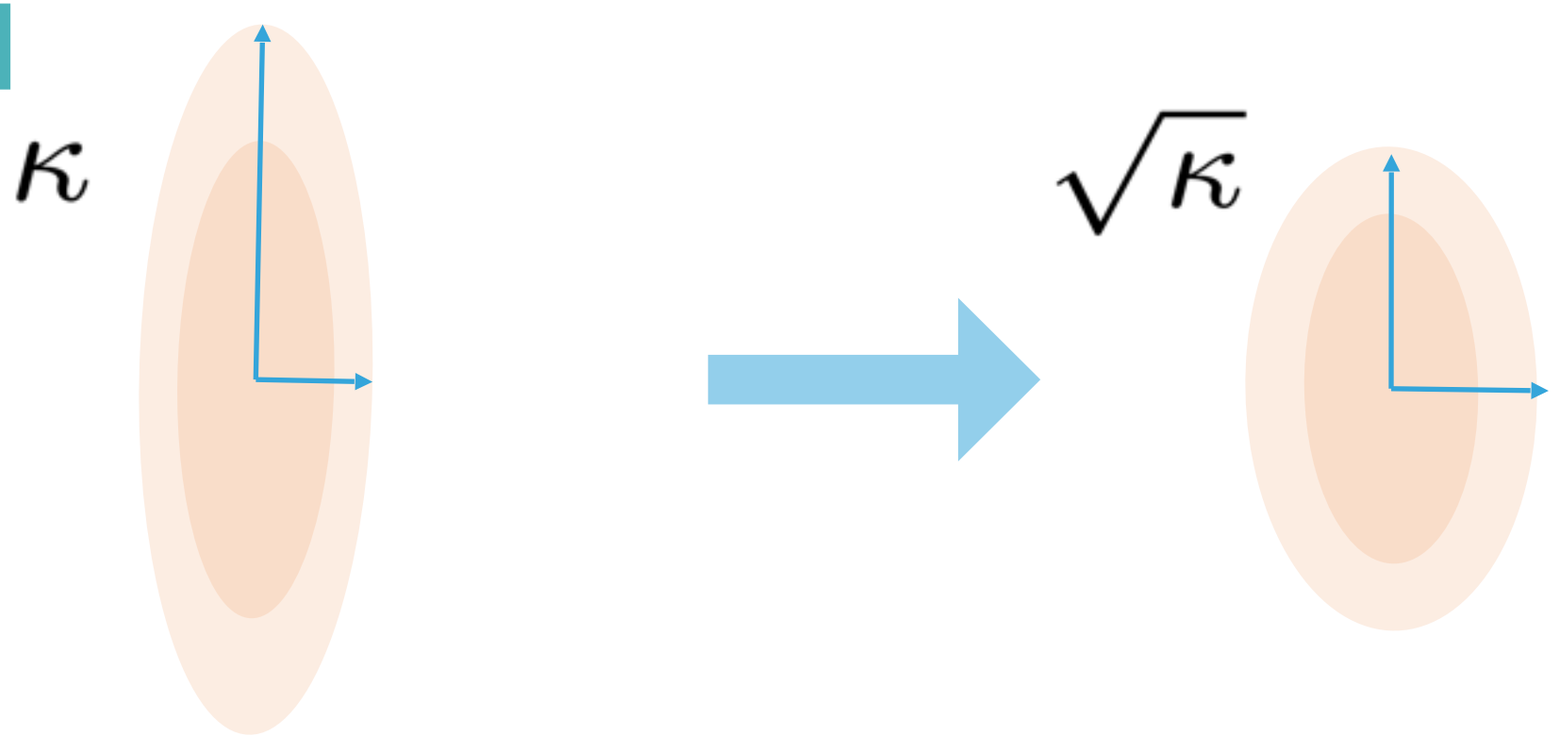


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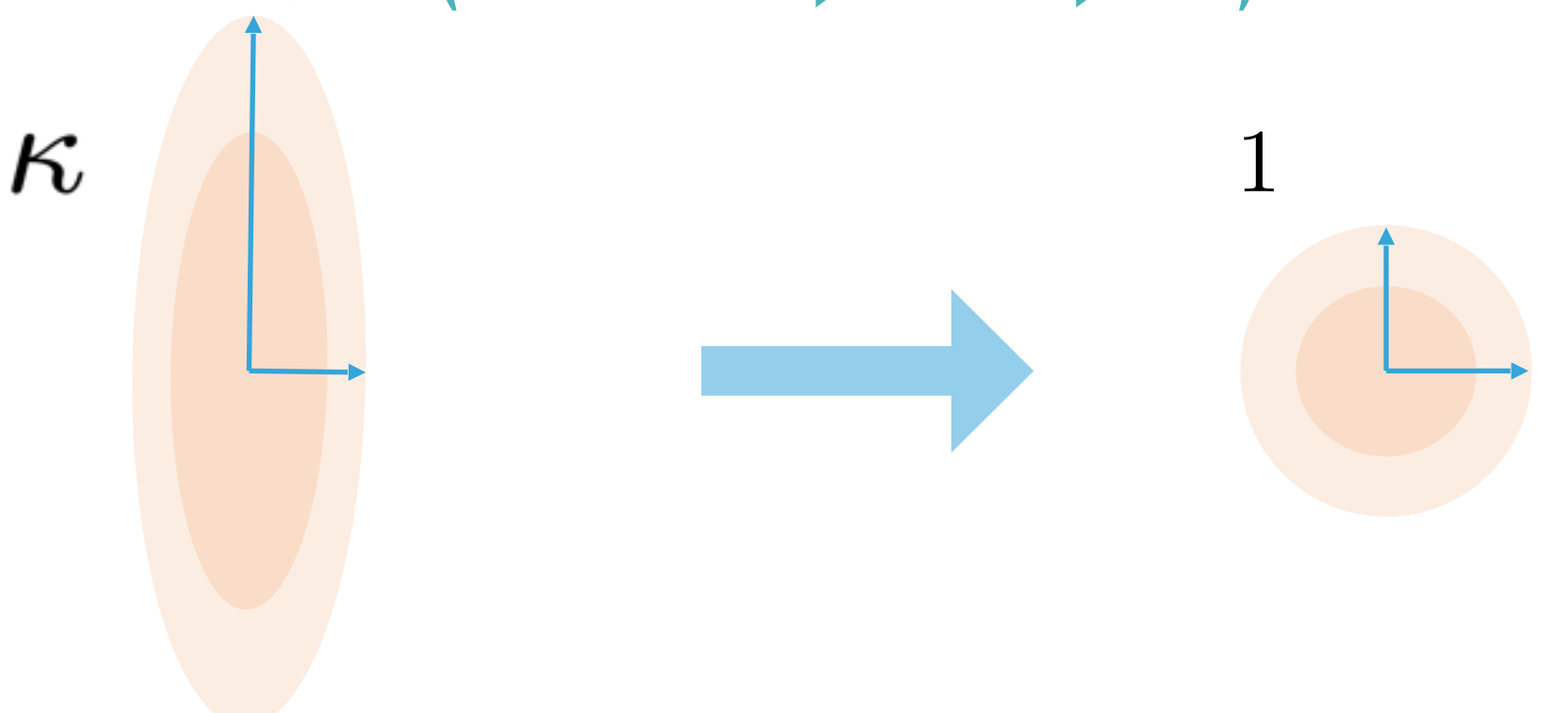
ACCELERATED STOCHASTIC POWER ITERATION



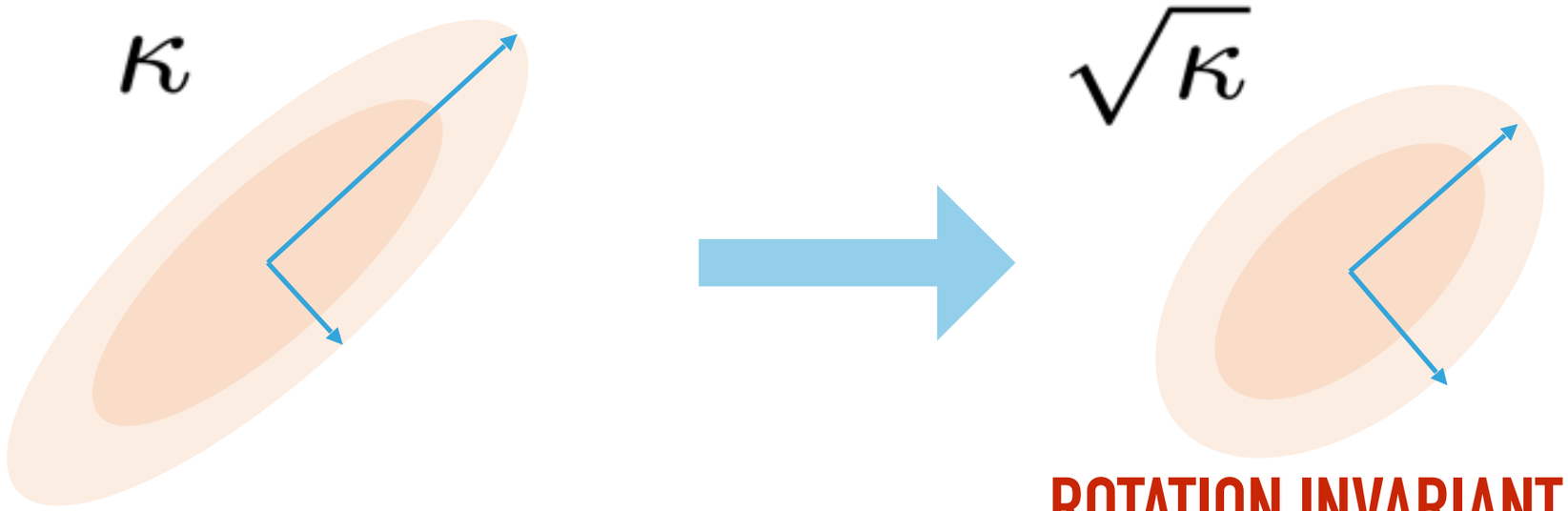
MOMENTUM



DIAGONAL ADAPTATION (ADAGRAD, ADAM, . . .)

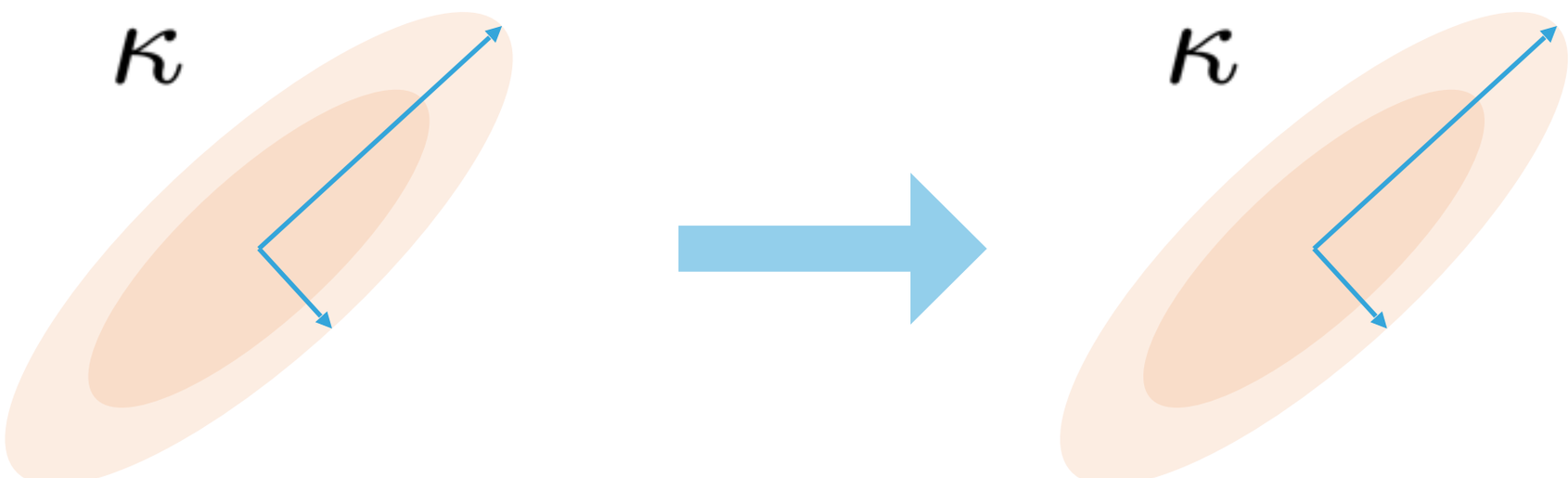


MOMENTUM



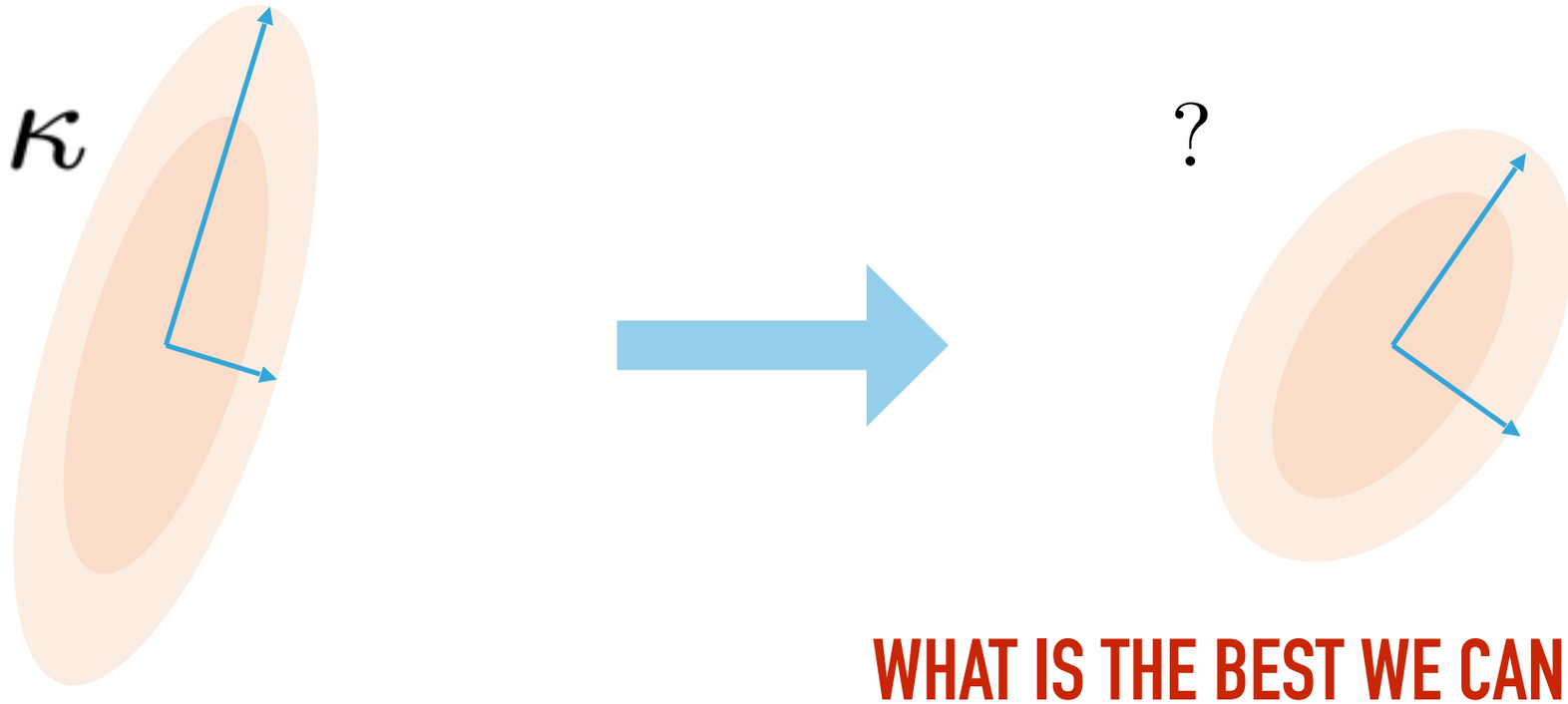
ROTATION INVARIANT

DIAGONAL ADAPTATION (ADAGRAD, ADAM, . . .)



NOT ROTATION INVARIANT

DIAGONAL ADAPTATION (ADAGRAD, ADAM, . . .)



WHAT IS THE BEST WE CAN DO WITH PER VARIABLE STEP SIZES?

$$D^* = \text{argmin}_{D \text{ diagonal}} \frac{\lambda_{max}(D^{1/2} H D^{1/2})}{\lambda_{min}(D^{1/2} H D^{1/2})}$$

EXPLORATION: OPTIMAL COMBINATION OF DIAGONAL ADAPTATION AND MOMENTUM